

LENGTH: say which spaces count

Check the value behind the display

01 / THE SAS EXAMPLE

```
data test;
  length text $10;
  input text $;
  len = length(text);
  datalines;
ABC
ABC
ABCDEFGHIJ
X
.
;
run;

proc print data=test noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026
Claude Sonnet 4.5 / AWS Bedrock GovCloud
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

Specify an invisible data detail and ask for a check that makes it visible.

Each text value occupies ten character positions. LENGTH ignores trailing blanks and returns 1 for all blanks. The period on the final input row supplies a missing character value.

EXPECTED RESULT

Row	Stored width	len
1	10	3
2	10	3
3	10	10
4	10	1
5	10	1

Both ABC rows contain seven padding spaces.
Final row: ten spaces; LENGTH returns 1.

REPORTED NUMERIC MATCHES: A 3/3 | B 3/3

Required behavior: A 0/3 | B 3/3

Reference values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.

Show code in this response. Do not create or run files, or read existing Python files.

REPORTED TRIAL RESULTS / 21 SEP 2026

Matching numbers; missing behavior

Run	Numeric values	Required behavior
A1	Yes	No
A2	Yes	No
A3	Yes	No

All three used pandas .str.len() directly.

Reported: no padding; final period kept as literal '.'.

The numeric vector [3, 3, 10, 1, 1] matched, but stored text and missing-value interpretation did not.

SOURCES: results-20260921.csv; trial-log-20260921.md; analysis-20260921.md

WHY CHOICES REMAIN OPEN

- 1

The numbers can agree
Literal '.' has length 1, which matches the expected blank result by coincidence.
- 2

Storage still differs
This fixture requires width 10. Correct numeric values alone do not establish stored-text equivalence.
- 3

Name the spaces to ignore
For these ASCII strings, ignore trailing ordinary spaces in the calculation; preserve the stored text.

COUNTEREXAMPLES / LOCALLY CHECKED

Python value	Direct len	Expected LENGTH
'ABC '	6	3
"	0	1
' ABC '	5	4

ANTI-PATTERN: CHECKING ONLY THE NUMBERS

Inspect stored values, missing input and trailing spaces, even when the numeric results agree.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas.
- Create text from ABC, ABC, ABCDEFGHIJ, X, and an empty string.
  The final SAS input period represents character missing here.
- Right-pad every text value with spaces to width 10.
- Preserve the stored text. For len, remove trailing ordinary
  spaces only with .str.rstrip(' '), then count characters.
- Match SAS LENGTH for a blank string: set its result to 1.
  Use .str.len().clip(lower=1) after the trailing-space removal.
- Print a visible representation of text and the len column.
- Include assertions: stored widths are all 10; len values are
  [3, 3, 10, 1, 1] in the original row order.
- Briefly explain why stored width and LENGTH can differ.
```

Show code in this response. Do not create or run files, or read existing Python files.

WHY THESE DETAILS HELP

- 1 **Make the stored value visible**
Right-pad to width 10 and inspect repr(text). Interpret the final input period as character missing.
- 2 **Preserve the counting rule**
Use rstrip(' ') for the calculation only. Keep leading spaces; set the all-blank result to 1.
- 3 **Verify the requested behavior**
B1-B3 report width and value assertions passing. Correct padding can use different methods.

REPORTED RESULTS / THREE B RUNS

Run	Numeric values	Required behavior
B1	Yes	Yes
B2	Yes	Yes
B3	Yes	Yes

CHECK IT YOURSELF
Confirm all widths = 10 and the five len values.
Final stored value: ten spaces, not a literal ' '.

SAS REFERENCE

Reports describe six fresh parallel agents; complete transcripts are absent from this repository.
Passing assertions are reported. Local reference checked; SAS run pending. Three runs do not guarantee future results.